

Lab: Analyzing Customer Behavior Dataset for a Retail Company

Problem Statement:

Retail companies collect vast amounts of customer data, including purchase patterns and demographic details. Analyzing this data helps businesses understand customer behavior, segment customers, and optimize marketing strategies. This lab focuses on exploring correlations, distributions, and relationships between different attributes using Pandas and visualization tools.

Objective:

- Load and explore a customer behavior dataset.
- Analyze purchase patterns based on demographics.
- Visualize correlations between different attributes.
- Identify trends and insights using Pandas and visualization tools.

Requirements:

- Python installed (version 3.x recommended).
- Pandas, NumPy, Matplotlib, and Seaborn installed (`pip install pandas numpy matplotlib seaborn`).
- Jupyter Notebook (optional but recommended for running the lab).
- Retail Sales Dataset from Kaggle: [Retail Sales Dataset](#).

Part 1: Importing Libraries and Loading Data

```
In [1]: # import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: # Load the dataset
df = pd.read_csv("Dataset/retail_sales_dataset.csv")

print("Dataset Loaded Successfully")
df.head()
```

Dataset Loaded Successfully

Out[2]:

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
4	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100

Part 2: Analyzing Demographic Trends

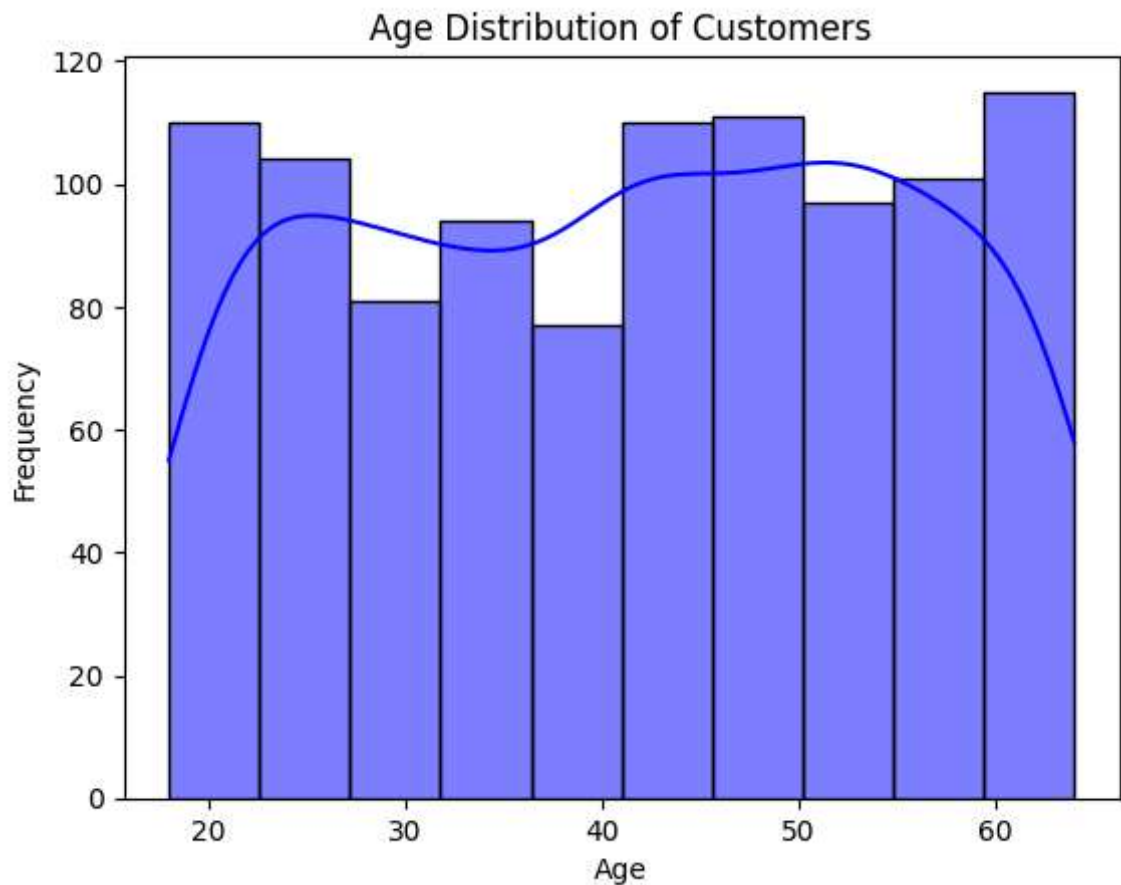
In [3]:

```
# Checking for missing values
print("Missing values per column:")
print(df.isnull().sum())
```

Missing values per column:
Transaction ID 0
Date 0
Customer ID 0
Gender 0
Age 0
Product Category 0
Quantity 0
Price per Unit 0
Total Amount 0
dtype: int64

In [12]:

```
# Distribution of Age
sns.histplot(df['Age'].dropna(), bins=10, kde=True, color='blue')
plt.title('Age Distribution of Customers')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```

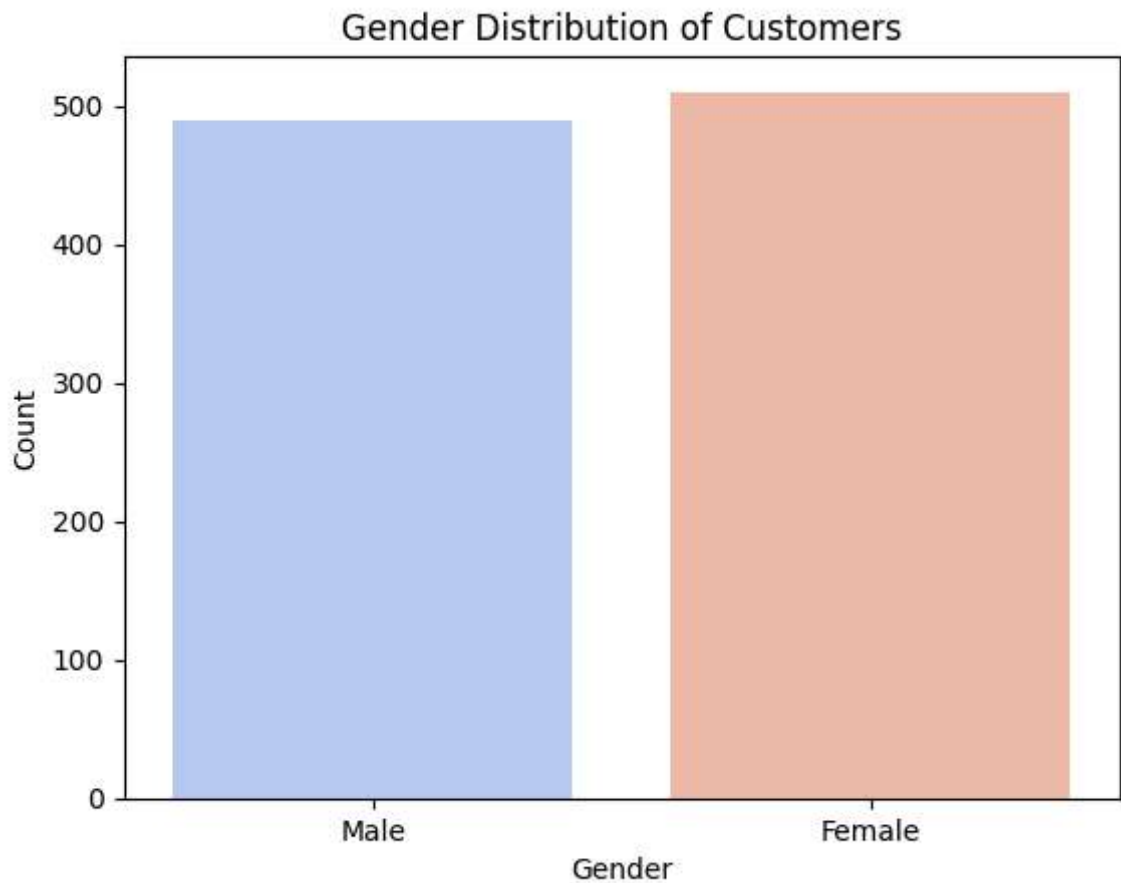


```
In [11]: # Gender Distribution
sns.countplot(x='Gender', data=df, palette='coolwarm')
plt.title('Gender Distribution of Customers')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()
```

C:\Users\bhawa\AppData\Local\Temp\ipykernel_27036\2522661910.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.countplot(x='Gender', data=df, palette='coolwarm')
```



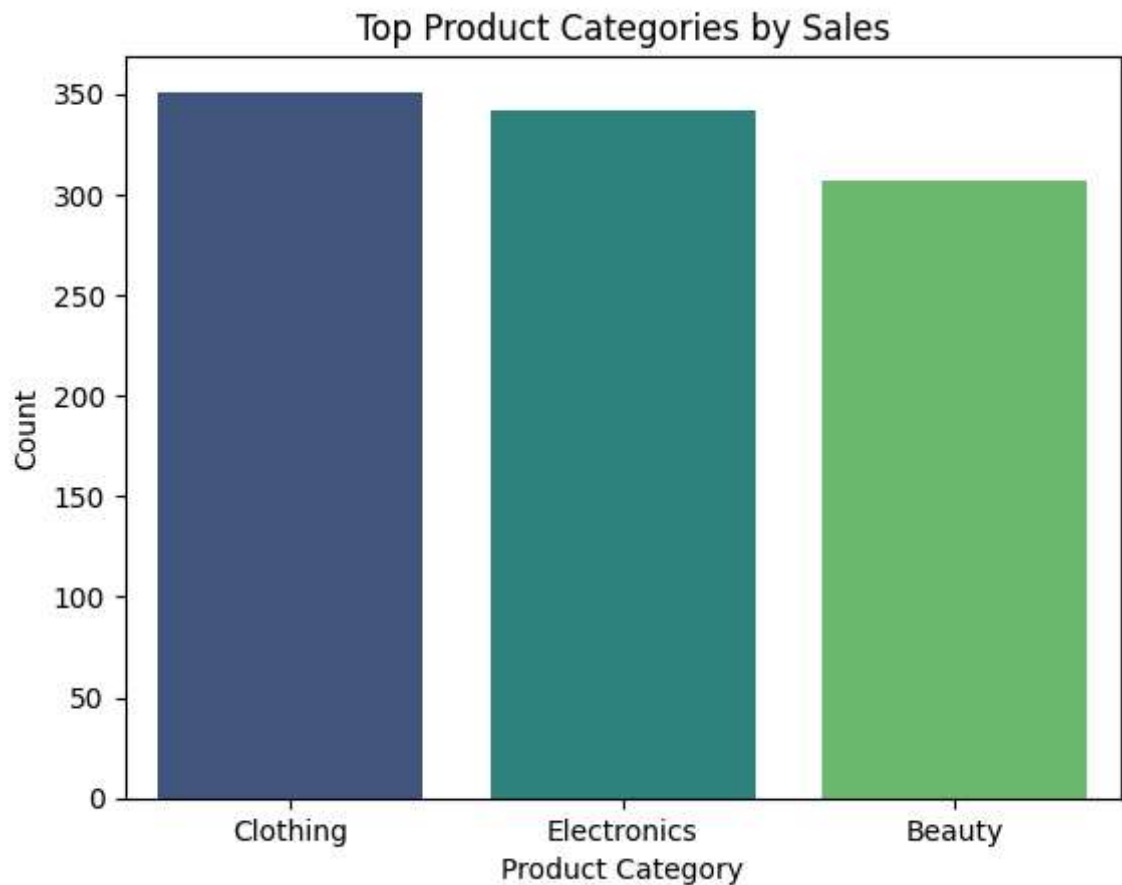
Part 3: Understanding Purchase Patterns

```
In [10]: # Top Product Categories by Sales
sns.barplot(x=df['Product Category'].value_counts().index, y=df['Product Category'].value_counts(), palette='viridis')
plt.title('Top Product Categories by Sales')
plt.xlabel('Product Category')
plt.ylabel('Count')
plt.show()
```

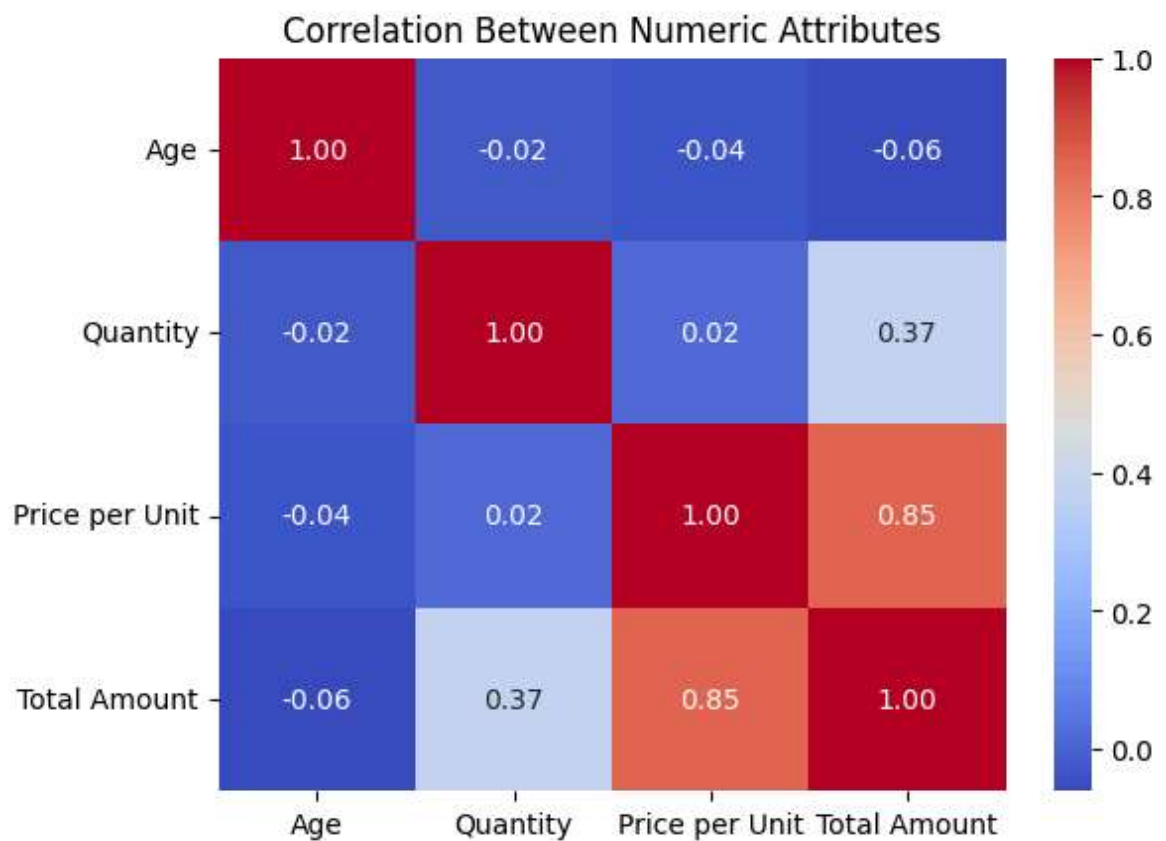
C:\Users\bhawa\AppData\Local\Temp\ipykernel_27036\950203032.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=df['Product Category'].value_counts().index, y=df['Product Category'].value_counts(), palette='viridis')
```



```
In [15]: # Correlation heatmap
sns.heatmap(df[['Age', 'Quantity', 'Price per Unit', 'Total Amount']].corr(), an
plt.title('Correlation Between Numeric Attributes')
plt.show()
```



Part 4: Segmenting Customers Based on Spending

```
In [16]: # Customer Segmentation Based on Spending Category
def categorize_spending(amount):
    if amount >= df['Total Amount'].quantile(0.75):
        return 'High'
    elif amount >= df['Total Amount'].quantile(0.50):
        return 'Medium'
    else:
        return 'Low'

df['Spending_Category'] = df['Total Amount'].apply(categorize_spending)
print("Customer Segmentation Based on Spending Category:")
df[['Customer ID', 'Total Amount', 'Spending_Category']].head()
```

Customer Segmentation Based on Spending Category:

```
Out[16]:
```

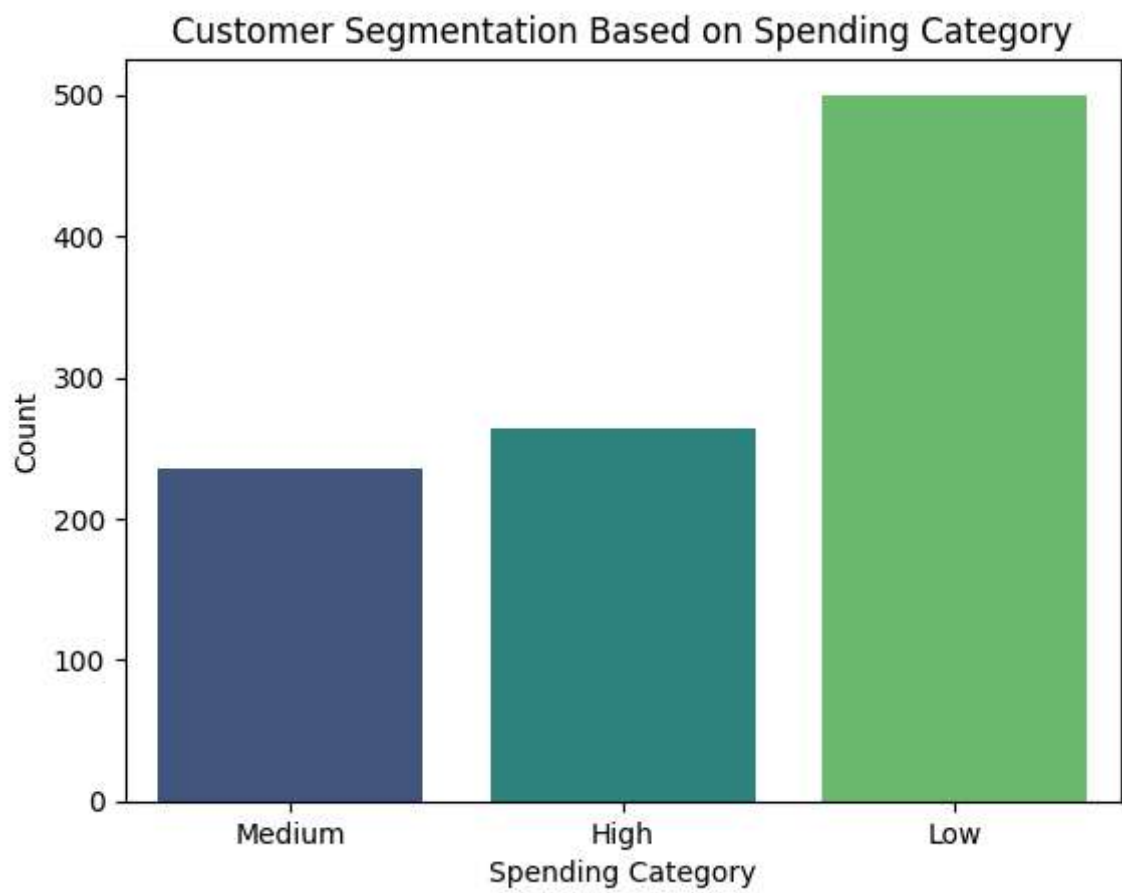
	Customer ID	Total Amount	Spending_Category
0	CUST001	150	Medium
1	CUST002	1000	High
2	CUST003	30	Low
3	CUST004	500	Medium
4	CUST005	100	Low

```
In [17]: # plot the spending category
sns.countplot(x='Spending_Category', data=df, palette='viridis')
plt.title('Customer Segmentation Based on Spending Category')
plt.xlabel('Spending Category')
plt.ylabel('Count')
plt.show()
```

C:\Users\bhawa\AppData\Local\Temp\ipykernel_27036\2254791853.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.countplot(x='Spending_Category', data=df, palette='viridis')
```



In []: